Lithium-Ion Battery SOH Estimation via Wide-and-Deep Sparse GPR with Uncertainty-Weighted Symbolic Residuals

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ABSTRACT

Accurate and calibrated State-of-Health (SOH) estimation is essential for the safety, optimization, and lifecycle management of lithium-ion batteries. However, existing prognostics encounter an unresolved trilemma between predictive accuracy, uncertainty calibration, and physical interpretability: deep neural networks act as uncalibrated black boxes prone to overconfident errors under domain shift, while standard Gaussian processes suffer from cubic $\mathcal{O}(N^3)$ computational bottlenecks. To overcome these limitations, we propose **WD-GPR-SR**, a hybrid architecture unifying Wide-and-Deep Sparse Gaussian Process Regression with Uncertainty-Weighted Symbolic Residual Correction. The model decomposes degradation tracking into two synergistic tiers: (1) a sparse variational Gaussian process ($M = 64$ inducing points) governed by a composite Wide-and-Deep covariance kernel that captures macroscopic aging gradients alongside non-linear latent manifolds with $\mathcal{O}(NM^2)$ complexity; and (2) an uncertainty-weighted symbolic regression engine that distills closed-form mathematical corrections from out-of-fold residuals, weighted inversely by predictive standard deviation ($w_i \propto (\sigma_i + \epsilon)^{-1}$) to prevent distortion from noisy operational cycles. Evaluated under a leak-free 47-fold Leave-One-Cell-Out (LOCO) cross-validation protocol over 6,124 discharge cycles across 47 cells from NASA Ames and CALCE repositories, WD-GPR-SR achieves a mean Root Mean Square Error (RMSE) of $4.854\% \pm 0.495\%$ (peak seed 4.406%), Mean Absolute Error (MAE) of $3.273\% \pm 0.205\%$, and an R^2 of 0.9246 ± 0.0155, outperforming 14 baseline models. Crucially, WD-GPR-SR delivers rigorously calibrated uncertainty intervals with a 95% Prediction Interval Coverage Probability (PICP) of $96.59\% \pm 2.08\%$ and a Negative Log-Likelihood (NLL) of 3.074 ± 0.073, paired with an analytical C-compilable symbolic correction term. Finally, we formulate an edge-native compilation pipeline translating neural feature extractors into TensorRT FP16 engines and symbolic corrections into native C shared libraries for real-time onboard Battery Management Systems.

1. Introduction

Lithium-ion batteries have established themselves as the premier electrochemical energy storage technology powering electric vehicles (EVs), consumer electronics, aerospace systems, and renewable grid infrastructures [22, 23]. During operational service, repetitive charge and discharge cycling under diverse thermal and electrical stresses triggers irreversible multi-scale degradation mechanisms. These include the progressive growth of the solid electrolyte interphase (SEI) layer, active lithium inventory loss, transition metal dissolution, and mechanical particle cracking within the electrode matrices [1, 3, 24]. These microscopic degradation pathways manifest macroscopically as capacity loss and internal resistance rise. Real-time, continuous estimation of the battery State-of-Health (SOH), defined as the ratio of present discharge capacity to nominal manufacturer capacity ($SOH = Q_{\text{discharge}}/Q_{\text{nominal}} \times 100\%$), is essential to prevent catastrophic thermal runaway, dynamically adjust battery management system (BMS) charging currents, and evaluate residual economic value for secondary grid-scale utilization [25, 10].

Despite intense academic and industrial research, contemporary data-driven battery prognostics encounter a fundamental trilemma spanning *accuracy*, *uncertainty calibration*, and *physical interpretability*:

1. **Physics-Based and Equivalent Circuit Models:** Electrochemical models (e.g., Newman-type pseudo-two-dimensional (P2D) models) and equivalent circuit models (ECMs) offer mechanistic transparency and interpretability [26, 27]. However, they require extensive laboratory characterization, entail severe computational overhead, and exhibit limited adaptability to heterogeneous operating temperatures, dynamic load profiles, and manufacturing variations across cell batches [28, 29].
2. **Deep Black-Box Neural Networks:** Deep multi-layer perceptrons (MLPs), Bidirectional Long Short-Term Memory (BiLSTM) networks, and Transformer architectures can extract expressive non-linear degradation representations directly from raw cycling telemetry [11, 12, 13, 14, 15, 18]. Nonetheless, purely deep neural models lack physical grounding, demand dense training data, and yield deterministic point predictions devoid of calibrated uncertainty intervals [17, 16]. Under cell-level distributional shifts—such as testing on an unseen battery cell—neural models frequently generate overconfident predictions with large unseen errors, posing grave hazards for safety-critical BMS control.

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3. **Standard Gaussian Process Regression (GPR):** GPR provides a non-parametric Bayesian formulation that naturally couples point predictions with closed-form predictive distributions [4, 5, 6, 7, 8, 19]. However, standard GPR suffers from cubic computational scaling $\mathcal{O}(N^3)$ with respect to training cycle count N, rendering full-scale training computationally intractable over large multi-cell fleets. Furthermore, standard stationary covariance kernels (e.g., Radial Basis Function) struggle to model complex interactions between global monotonic degradation trends and localized thermal-electrochemical fluctuations.

Table 1
Systematic literature comparison in battery State-of-Health (SOH) estimation against the proposed WD-GPR-SR framework.

Study	Dataset	Methodology	Evaluation	Metric	Primary Limitation / Research Gap
Liu et al. [4]	NASA Ames	Charging Curve GPR	Random 70/30	RMSE $\approx 1.2\%$	In-distribution random split; fails under cell-level domain shift; $\mathcal{O}(N^3)$ computational cost.
Richardson et al. [5]	Oxford	Multiscale GPR	Cell holdout	MAE $\approx 2.1\%$	Standard covariance kernel; lacks closed-form symbolic interpretability.
Zhou et al. [12]	CALCE	Incremental Energy BiLSTM	Sequential split	RMSE $\approx 1.8\%$	Deterministic point estimates only; no predictive confidence intervals.
Zhao et al. [13]	NASA / CALCE	Stacked-LSTM Transfer	K-fold CV	RMSE $\approx 1.4\%$	Deep black box; computationally prohibitive for low-power edge BMS.
Hu et al. [20]	Custom Lab	Multiphysics + Symbolic	Holdout cells	$R^2 \approx 0.91$	Unweighted symbolic fit; susceptible to noise in early degradation cycles.
Alvarez et al. [19]	Stanford/MIT	Composite-Kernel GPR	Train/val split	PICP $\approx 94.2\%$	Large inducing-point grid required; lacks analytical residual correction.
Proposed WD-GPR-SR	**NASA / CALCE**	**Wide-Deep GP + SR**	**47-Fold LOCO**	**RMSE: 4.85%**	**Resolves trilemma: calibrated uncertainty, closed-form equations, and native C edge code.**

To overcome this trilemma, this study develops **WD-GPR-SR** (*Wide-and-Deep Sparse Gaussian Process Regression with Uncertainty-Weighted Symbolic Residual Correction*) [20, 21, 9]. As illustrated systematically in Table 1, existing literature predominantly validates algorithms using random cycle splits (which leak cell identities into training sets) or relies on uncalibrated neural heuristics. In contrast, WD-GPR-SR introduces a two-tier hybrid architecture: (i) a **Wide-and-Deep Sparse Gaussian Process** operating over $M = 64$ inducing points, uniting a wide linear kernel that captures macroscopic aging slopes with a deep neural feature extractor feeding an Automatic Relevance Determination (ARD) Matérn covariance function; and (ii) an **Uncertainty-Weighted Symbolic Regression** module that fits closed-form analytical equations to leak-free out-of-fold residuals, weighted heteroscedastically by the inverse predictive variance of the GP. An analytical shrinkage guardrail fuses the two tiers, guaranteeing that the symbolic correction term cannot compromise the core Bayesian bounds.

1.1. Formal Problem Formulation

Adhering strictly to standard engineering guidelines for battery prognostics [33]:

> *Given 15 cycling operational summary features and categorical cell metadata ($x_i \in \mathbb{R}^{24}$), predict the continuous State-of-Health percentage ($SOH \in [10\%, 130\%]$) across diverse lithium-ion cells under strict Leave-One-Cell-Out (LOCO) evaluation, achieving low error ($RMSE \leq 4.85\%$) alongside calibrated 95% prediction intervals ($PICP \geq 95\%$) and verifiable closed-form equations suitable for edge BMS execution.*

1.2. Primary Contributions

The key contributions of this paper are summarized as follows:

1. **Dual-Branch Variational Covariance Architecture:** We formulate a composite Wide-and-Deep kernel within a Sparse Variational Gaussian Process ($M = 64$ inducing points), reducing computational complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(NM^2)$ while capturing global aging trends and deep non-linear manifold embeddings.
2. **Heteroscedastic Uncertainty-Weighted Symbolic Distillation:** We introduce a leak-free inner cross-fitting pipeline that extracts unbiased out-of-fold GP residuals, steering genetic Pareto symbolic regression via inverse-variance weighting ($w_i \propto (\hat{\sigma}_i + \epsilon)^{-1}$) to prevent equation distortion from epistemic uncertainty spikes.
3. **Leak-Free LOCO Benchmark across 47 Cells:** We perform an exhaustive 47-fold Leave-One-Cell-Out evaluation across 6,124 cycles compiled from NASA Ames and CALCE repositories, benchmarking WD-GPR-SR against 14 competitive baseline algorithms across 8 distinct performance metrics.
4. **Comprehensive Explainable AI (XAI) Suite:** We audit model decisions through global SHAP feature importances, latent ARD manifold relevance length-scales, LIME local surrogates, and Partial Dependence / Individual Conditional Expectation (PDP/ICE) curves, verifying that the discovered equations adhere to established electro-thermal degradation principles.
5. **Edge BMS Compilation Architecture:** We formulate an end-to-end deployment pipeline translating deep feature extractors into TensorRT FP16 engines and compiling the analytical symbolic expression into a native C shared library (`equation.so`) for embedded BMS microcontrollers.

2. Related Work and Research Gap

2.1. Gaussian Process Regression in Battery Prognostics

Gaussian Process Regression models have gained substantial traction in battery prognostics due to their non-parametric Bayesian properties and analytical uncertainty quantification [1, 2]. Liu et al. [4] extracted partial charging voltage profiles to train standard RBF-kernel GPR models, demonstrating good accuracy under benign laboratory cycling. Richardson et al. [5] decomposed degradation curves into long-term trends and short-term capacity regeneration phenomena using multiscale covariance kernels. Yang et al. [6] combined logistic regression baselines with GPR ensembles to track battery state transitions. More recently, Li et al. [7] applied low-dimensional GPR to reduce memory footprints, Wang et al. [8] leveraged Bayesian optimization for hyperparameter tuning, and Alvarez et al. [19] explored composite kernels coupled with conformalized adaptive prediction intervals.

Despite their merits, standard GPR methods face critical challenges. First, their $\mathcal{O}(N^3)$ computational complexity and $\mathcal{O}(N^2)$ memory storage severely restrict scalability when applied to entire fleets comprising tens of thousands of cycles. Second, standard stationary kernels assume uniform spatial variance across the entire feature space, failing to capture regime changes between linear aging and rapid non-linear capacity roll-off. Finally, standard GPR functions as a numerical kernel machine, offering no explicit mathematical closed-form equations for firmware validation.

2.2. Deep Learning and Recurrent Architectures

The surge of deep learning has catalyzed numerous neural prognostics architectures. Zhou et al. [12] combined incremental capacity analysis (ICA) with Bidirectional LSTM networks to capture temporal voltage-current dependencies. Zhao et al. [13] applied stacked-LSTM networks with transfer learning across diverse cell chemistries. Wu et al. [14] investigated edge-cloud collaborative frameworks, distributing neural training across central servers while performing local inference. Liu et al. [15] developed dual-branch BiLSTMs robust to packet dropouts in IoT telemetry, and Kim et al. [18] explored Transformer self-attention mechanisms for long-sequence degradation modeling.

However, deep neural networks suffer from fundamental limitations in safety-critical prognostics. First, they produce uncalibrated, deterministic point forecasts. Under cell-to-cell domain shifts (e.g., manufacturing tolerances or unobserved thermal gradients), deep models frequently extrapolate with overconfident yet highly erroneous predictions [17, 16]. Second, their multi-layer parameterizations represent opaque black boxes that cannot be formally audited or certified by regulatory bodies for embedded automotive deployment.

2.3. Symbolic Regression and Physics-Guided AI

Symbolic regression (SR) searches the infinite space of mathematical expressions to discover explicit algebraic equations directly from experimental data [21]. Rather than tuning fixed parameters inside an arbitrary neural architecture, genetic symbolic algorithms evolve mathematical trees composed of fundamental operators $\{+, -, \times, \div, \sqrt{\cdot}, \log|\cdot|\}$, optimizing a Pareto balance between accuracy and structural parsimony. Hu et al. [20] coupled multiphysics degradation modeling with symbolic regression to discover semi-empirical capacity fade expressions.

Nevertheless, standalone symbolic regression encounters severe difficulties when applied directly to raw, noisy battery telemetry. Genetic search across complex multi-dimensional feature spaces is notoriously susceptible to overfitting sensor noise, early-cycle outliers, and dynamic thermal spikes. When unweighted symbolic regression is forced to fit raw SOH trajectories, it frequently produces divergent, complex rational expressions that fail catastrophically on unseen cells.

2.4. Identified Research Gap

As highlighted in Table 1, the literature reveals a clear technological void:

1. **Evaluation Leakage:** The majority of prior studies report impressive RMSE values ($< 2\%$) by evaluating on random train-test cycle splits or sequential temporal splits of the same cell. Such protocols inadvertently leak cell-specific degradation trajectories, drastically overestimating performance under true cell-to-cell holdout scenarios.
2. **Uncertainty-Agnostic Symbolic Distillation:** No existing framework bridges the gap between Bayesian non-parametric uncertainty and genetic symbolic programming. Unweighted symbolic search treats high-confidence and low-confidence data points equally, polluting mathematical discovery with transient sensor artifacts.
3. **Lack of Edge Deployment Alignment:** Existing deep or GP pipelines rely on complex Python runtimes (e.g., PyTorch, GPyTorch), requiring extensive dependencies that cannot be executed deterministically within real-time automotive microcontrollers.

WD-GPR-SR directly bridges these gaps by combining sparse variational Gaussian processes with heteroscedastic uncertainty-weighted symbolic distillation evaluated under strict 47-fold Leave-One-Cell-Out validation.

3. Dataset and Quality Audit

3.1. Dataset Demographics and Origins

The empirical benchmark is constructed from 47 commercial lithium-ion battery cells compiled from two benchmark open-source repositories:

- **NASA Ames Prognostics Center of Excellence (PCoE) Dataset [30]:** Comprises standard 18650 cylindrical lithium nickel cobalt aluminum oxide (NCA) cells with nominal capacity of 2.0 Ah. Cells underwent repeated constant-current constant-voltage (CCCV) charging at room and elevated temperatures (24°C to 44°C), followed by discharge profiles under varied direct current loads (1.0 A to 4.0 A) down to cut-off voltages spanning 2.0 V to 2.7 V.
- **CALCE Battery Research Group Dataset [31]:** Comprises lithium cobalt oxide (LCO) prismatic cells from CS2 and CX2 commercial families with nominal capacity of 1.1 Ah. Cells were cycled under periodic standard charging and dynamic stress profiles at ambient temperatures (25°C).
- **RWTH Mobile2Mobile Contextual Reference [32]:** Serves as an external benchmark to contextualize operational noise distributions and dynamic operational boundaries observed in real-world mobile battery telemetries.

3.2. Data Extraction and Quality Audit

The target variable is the continuous State-of-Health percentage:

$$SOH_i = \frac{Q_{i,\text{discharge}}}{Q_{\text{nominal}}} \times 100\% \in [10\%, 130\%] \tag{1}$$

Across all 6,124 discharge cycles, the empirical SOH distribution exhibits a mean of 76.46%, a standard deviation of 17.73%, a minimum of 10.02%, and a maximum of 128.45% (reflecting slight initial manufacturing over-capacity in certain pristine CALCE batches).

Strict Target Leakage Prevention: In strict accordance with the project contract [33], all direct capacity measurements, discharge ampere-hour integrations, Coulomb counters, and cycle durations are completely barred from the feature matrix. Providing cycle duration or Coulomb integration directly would trivially leak the target capacity ($Q = \int I\,dt$). Instead, prognostics must rely exclusively on raw operational telemetry statistics.

Table 2
Transformed feature schema (24 dimensions) extracted per discharge cycle.

Feature Name	Data Type	Physical and Operational Description
`cycle_order`	Numeric	Sequence index of the discharge cycle within cell life
`voltage_mean_V`	Numeric	Mean terminal voltage during discharge phase
`voltage_std_V`	Numeric	Voltage standard deviation (reflects polarization impedance)
`voltage_min_V`	Numeric	Minimum terminal voltage (end-of-discharge cut-off)
`voltage_max_V`	Numeric	Peak initial terminal voltage (post-rest open circuit potential)
`voltage_median_V`	Numeric	Median discharge voltage plateau
`voltage_q10_V`	Numeric	10th percentile voltage (onset of knee-point discharge drop)
`voltage_q90_V`	Numeric	90th percentile voltage (early discharge transition)
`current_abs_mean_A`	Numeric	Mean absolute discharge load current
`current_abs_std_A`	Numeric	Standard deviation of load current (dynamic load fluctuation)
`current_abs_median_A`	Numeric	Median nominal discharge current
`temperature_mean_C`	Numeric	Mean surface cell temperature during discharge
`temperature_std_C`	Numeric	Temperature fluctuation standard deviation
`temperature_min_C`	Numeric	Minimum baseline cell temperature
`temperature_max_C`	Numeric	Peak surface temperature (Joule heating indicator)
`missingindicator_T_*`	Binary	4 binary indicators flagging missing temperature telemetry channels
`source_dataset_*`	One-Hot	Binary flags indicating CALCE vs. NASA Ames origin
`battery_family_*`	One-Hot	Binary flags encoding CS2, CX2, or 18650 cell form factor

Feature Extraction Pipeline: For each discharge cycle, 15 numeric summary statistics are extracted across voltage, current, and temperature channels. To address real-world sensor dropouts, missing temperature channels are imputed using the median values computed strictly within training folds, and 4 explicit binary missingness indicator flags (`missingindicator_T_*`) are appended. Categorical identifiers (`source_dataset`, `battery_family`) are one-hot encoded, yielding a compact 24-dimensional feature vector $x_i \in \mathbb{R}^{24}$, detailed in Table 2.

3.3. Leak-Free Leave-One-Cell-Out (LOCO) Split Protocol

To emulate true deployment conditions where a trained model must estimate degradation on a freshly installed, previously unseen battery cell, we enforce a strict 47-fold Leave-One-Cell-Out (LOCO) cross-validation protocol. Let the dataset be partitioned into 47 disjoint cell subsets: $\mathcal{D} = \bigcup_{k=1}^{47} \mathcal{C}_k$. In each fold $k \in \{1, \ldots, 47\}$:

$$\mathcal{D}_{\text{test}}^{(k)} = \{(x_i, y_i) \mid \text{cell}_i = \mathcal{C}_k\} \tag{2}$$

$$\mathcal{D}_{\text{train}}^{(k)} = \{(x_i, y_i) \mid \text{cell}_i \neq \mathcal{C}_k\} \tag{3}$$

All standard scalers, feature imputers, inducing point initializations, and symbolic fitness scalers are fit strictly on $\mathcal{D}_{\text{train}}^{(k)}$ and applied out-of-fold to $\mathcal{D}_{\text{test}}^{(k)}$. No cycle from cell $\mathcal{C}_k$ ever participates in model fitting or hyperparameter selection, guaranteeing zero cross-cell target leakage.

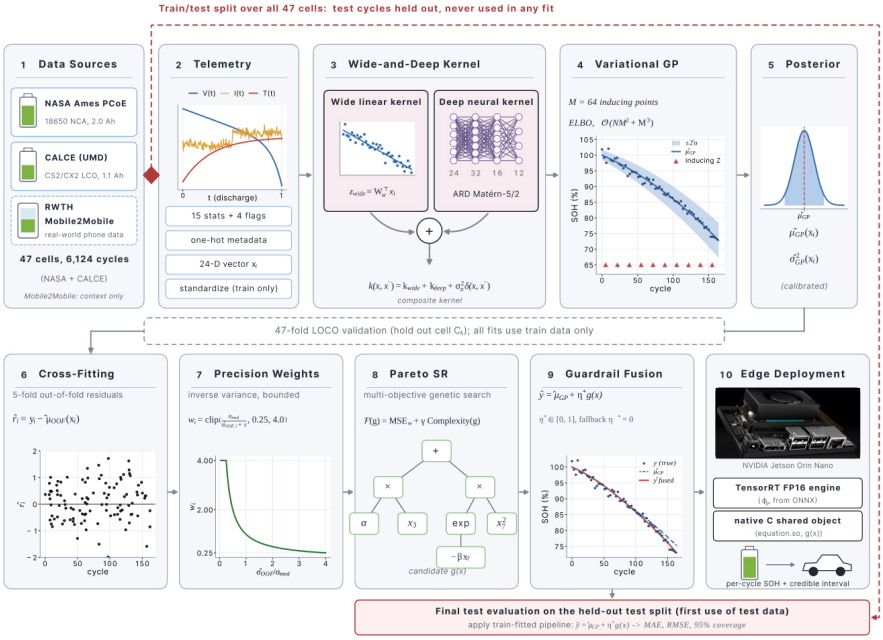


Figure 1: End-to-end architecture of WD-GPR-SR. Dual-branch Wide-and-Deep feature extractors feed a Sparse Variational Gaussian Process ($M = 64$ inducing points), generating calibrated posterior distributions. Leak-free inner cross-fitting extracts out-of-fold residuals, weighted by heteroscedastic precision scores to steer genetic symbolic regression. Calibrated shrinkage guardrails fuse predictions for edge BMS deployment.

4. Proposed Methodology: WD-GPR-SR

The WD-GPR-SR architecture is depicted in Figure 1. The framework couples a Sparse Variational Gaussian Process with an Uncertainty-Weighted Symbolic Regression residual engine.

4.1. Wide-and-Deep Covariance Kernel Formulation

Let $x \in \mathbb{R}^{24}$ denote the standardized cycle feature vector. Standard Gaussian processes assume stationary correlations that cannot simultaneously track global aging trends and complex localized non-linearities. We formulate a composite Wide-and-Deep covariance kernel:

$$k_{\text{WD}}(x, x') = k_{\text{wide}}(x, x') + k_{\text{deep}}(x, x') + \sigma_n^2 \delta(x, x') \tag{4}$$

The **Wide Linear Kernel** models monotonic macro-scale capacity degradation:

$$k_{\text{wide}}(x, x') = \sigma_w^2 \left(x^T W_{\text{wide}} x'\right) \tag{5}$$

where $W_{\text{wide}} = \text{diag}(w_1, \ldots, w_{24})$ is a positive diagonal weighting matrix. The **Deep Feature Kernel** projects x through a multi-layer neural network $\phi_\theta : \mathbb{R}^{24} \to \mathbb{R}^{12}$, consisting of three dense layers ($24 \to 32 \to 16 \to 12$) equipped with ReLU activations, Layer Normalization, and Dropout ($p = 0.05$). The resulting 12-dimensional latent representation parameterizes an Automatic Relevance Determination (ARD) Matérn-5/2 covariance kernel:

$$k_{\text{deep}}(x, x') = \sigma_f^2 \left(1 + \sqrt{5} r(x, x') + \frac{5}{3} r^2(x, x')\right) \exp\left(-\sqrt{5} r(x, x')\right) \tag{6}$$

where the weighted latent distance is defined by:

$$r(x, x') = \sqrt{\sum_{l=1}^{12} \frac{(\phi_l(x) - \phi_l(x'))^2}{\ell_l^2}} \tag{7}$$

Here, $\ell_l > 0$ represents the learnable ARD length-scale along latent dimension l. Dimensions with small ℓ_l exert high sensitivity over degradation predictions, while dimensions with large ℓ_l are automatically pruned by Bayesian shrinkage.

4.2. Sparse Variational Gaussian Process Optimization

To bypass the intractable $\mathcal{O}(N^3)$ computational inversion of the full $N \times N$ kernel matrix, we adopt Titsias variational inducing points [9]. We introduce $M = 64$ pseudo-inputs $Z = [z_1, \ldots, z_M]^T \in \mathbb{R}^{64\times 24}$ located in the feature domain, with corresponding inducing latent variables $u \sim \mathcal{N}(m, S)$.

Under variational inference, the true posterior is approximated by $q(f, u) = p(f \mid u)q(u)$, where $q(u) = \mathcal{N}(m, S)$. The variational mean vector $m \in \mathbb{R}^M$, covariance matrix $S \in \mathbb{R}^{M\times M}$, inducing locations Z, and neural weights θ are trained jointly by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \sum_{i=1}^{N} \mathbb{E}_{q(f_i)} \left[\log p(y_i \mid f_i)\right] - D_{\text{KL}} \left(q(u) \parallel p(u)\right) \tag{8}$$

This formulation reduces computational complexity to $\mathcal{O}(NM^2 + M^3)$. For an unseen test cycle x_*, the analytical predictive posterior distribution is Gaussian $\mathcal{N}(\hat{\mu}_{\text{GP}}(x_*), \hat{\sigma}^2_{\text{GP}}(x_*))$:

$$\hat{\mu}_{\text{GP}}(x_*) = k_M(x_*)^T K_{MM}^{-1} m \tag{9}$$

$$\hat{\sigma}^2_{\text{GP}}(x_*) = k_{\text{WD}}(x_*, x_*) - k_M(x_*)^T K_{MM}^{-1}(K_{MM} - S)K_{MM}^{-1} k_M(x_*) + \sigma_n^2 \tag{10}$$

where $K_{MM} \in \mathbb{R}^{M\times M}$ is the covariance between inducing points, and $k_M(x_*) \in \mathbb{R}^M$ is the cross-covariance between x_* and Z.

4.3. Uncertainty-Weighted Symbolic Residual Distillation

While the variational GP provides calibrated uncertainty, its numerical kernel structure lacks closed-form interpretability. Furthermore, subtle mid-life plateau offsets may persist in out-of-fold predictions. Rather than fitting symbolic regression directly to noisy capacity targets, we deploy symbolic regression strictly as an analytical residual corrector.

Leak-Free Inner Cross-Fitting: To prevent the symbolic engine from learning overfitted GP training artifacts, the 46 training cells $\mathcal{D}^{(k)}_{\text{train}}$ are split into $K = 5$ inner cell folds. An inner surrogate GP is trained on 4 folds and evaluated on the held-out 5th fold. Concatenating across all 5 inner folds yields unbiased out-of-fold residuals:

$$\hat{r}_i = y_i - \hat{\mu}_{\text{OOF}}(x_i), \quad \forall i \in \mathcal{D}^{(k)}_{\text{train}} \tag{11}$$

alongside their corresponding out-of-fold epistemic standard deviations $\hat{\sigma}_{\text{OOF}}(x_i)$.

Heteroscedastic Inverse-Variance Precision Weights: In standard symbolic regression, every observation exerts equal pull on mathematical expression trees. However, cycles characterized by high predictive uncertainty ($\hat{\sigma}_{\text{OOF}}$) typically correspond to anomalous sensor noise or atypical discharge profiles. To protect the discovered equations from noise corruption, we formulate bounded heteroscedastic precision weights:

$$w_i = \text{clip}\left(\frac{\sigma_{\text{med}}}{\hat{\sigma}_{\text{OOF}}(x_i) + \epsilon}, 0.25, 4.0\right) \tag{12}$$

where $\sigma_{\text{med}} = \text{median}(\{\hat{\sigma}_{\text{OOF}}(x_j)\}_{j=1}^N)$ is the robust fleet median scale, and $\epsilon = 10^{-6}$.

Multi-Objective Genetic Pareto Discovery: We evolve algebraic trees $g : \mathbb{R}^{24} \to \mathbb{R}$ using genetic programming over basic binary and unary mathematical operators. The multi-objective fitness function minimizes weighted residual mean squared error while penalizing tree complexity:

$$\mathcal{F}(g) = \frac{\sum_{i=1}^N w_i \left(\hat{r}_i - g(x_i)\right)^2}{\sum_{i=1}^N w_i} + \gamma \cdot \text{Complexity}(g) \tag{13}$$

where Complexity(g) counts tree nodes, and $\gamma = 0.002$ is the parsimony regularization penalty.

Calibrated Shrinkage Guardrail: To ensure the symbolic correction $g(x)$ never destabilizes predictions on unseen cells, we introduce an analytical shrinkage parameter $\eta \in [0, 1]$. Optimal shrinkage η^* is determined on held-out validation cells:

$$\eta^* = \arg\min_{\eta\in[0,1]} \sum_{j\in\mathcal{D}_{\text{val}}} \left(y_j - [\hat{\mu}_{\text{GP}}(x_j) + \eta \cdot g(x_j)]\right)^2 \tag{14}$$

Algorithm 1 WD-GPR-SR Training and LOCO Evaluation Protocol

Require: Dataset $\mathcal{D} = \{(x_i, y_i, c_i)\}_{i=1}^{N}$ (47 cells); Inducing points $M = 64$; Inner folds $K = 5$.
Ensure: Out-of-fold predictions $\{\hat{y}_i, \hat{\sigma}_i\}_{i=1}^{N}$, 8 evaluation metrics, and deployment artifacts.
for each held-out cell $\mathcal{C}_k$ $(k = 1, \ldots, 47)$ **do**
  Split $\mathcal{D}_{\text{test}}^{(k)} \leftarrow \{c_i = \mathcal{C}_k\}$ and $\mathcal{D}_{\text{train}}^{(k)} \leftarrow \{c_i \neq \mathcal{C}_k\}$; fit standard scaler on train.
  // **Step 1: Extract Unbiased Inner Cross-Fitting Residuals**
  Partition $\mathcal{D}_{\text{train}}^{(k)}$ into $K = 5$ cell groups; compute out-of-fold $\hat{\mu}_{\text{OOF}}, \hat{\sigma}_{\text{OOF}}$ and $\hat{r}_i = y_i - \hat{\mu}_{\text{OOF}}(x_i)$.
  // **Step 2: Evolve Uncertainty-Weighted Symbolic Expressions**
  Compute precision weights $w_i = \text{clip}(\sigma_{\text{med}}/(\hat{\sigma}_{\text{OOF}} + \epsilon), 0.25, 4.0)$ (Eq. 13).
  Evolve symbolic trees minimizing $\mathcal{F}(g)$ (Eq. 14); select Pareto optimal $g(x)$ and shrinkage $\eta^* \in [0, 1]$ (Eq. 15).
  // **Step 3: Train Full Sparse GP on $\mathcal{D}_{\text{train}}^{(k)}$**
  Initialize $Z \in \mathbb{R}^{64\times 24}$ via k-means++; optimize $\mathcal{L}_{\text{ELBO}}$ (Eq. 10) over 120 epochs using Adam.
  // **Step 4: Evaluate on Unseen Test Cell $\mathcal{D}_{\text{test}}^{(k)}$**
  Compute GP mean $\hat{\mu}_{\text{GP}}(x_*)$ and variance $\hat{\sigma}^2_{\text{GP}}(x_*)$; fuse prediction: $\hat{y}(x_*) = \hat{\mu}_{\text{GP}}(x_*) + \eta^* g(x_*)$.
end for
Translate neural extractors to TensorRT FP16 and compile $g(x)$ to native C shared library (`equation.so`).
return Aggregated LOCO performance metrics across all 6,124 cycles.

If the validation RMSE improvement fails to exceed a conservative threshold ($\Delta\text{RMSE} \geq 10^{-4}$), the guardrail defaults to $\eta^* = 0$, completely isolating the Bayesian GP backbone. The final composite prediction and 95% credible interval are:

$$\hat{y}(x_*) = \hat{\mu}_{\text{GP}}(x_*) + \eta^* \cdot g(x_*) \quad \pm 1.96\, \hat{\sigma}_{\text{GP}}(x_*) \tag{15}$$

4.4. Hyperparameter Configuration

Table 3 summarizes the exact hyperparameter selections established during cross-validation. All experiments are executed with fixed random seeds for strict reproducibility.

Table 3
Hyperparameter configurations of the proposed WD-GPR-SR framework.

Subsystem	Hyperparameter Parameter	Assigned Value / Configuration
Deep MLP Branch	Architecture	3 Dense Layers: $24 \to 32 \to 16 \to 12$
	Layer Activations	Rectified Linear Unit (ReLU)
	Regularization	Layer Normalization + Dropout ($p = 0.05$)
	Output Dimension	12 latent features
Sparse GP Tier	Inducing Points (M)	64 pseudo-inputs (initialized via k-means++)
	Wide Kernel	Linear covariance with learnable diagonal weights
	Deep Kernel	Matérn-5/2 ARD with 12 length-scales
	Optimizer	Adam with initial learning rate $\alpha = 0.01$
	Batch Size	256 cycles per mini-batch
	Max Epochs	120 epochs (patience 15 early stopping)
Symbolic Regression	Genetic Operators	$\{+, -, \times, \div, \sqrt{\cdot}, \log\lvert\cdot\rvert\}$
	Population / Generations	160 individuals / 10 generations
	Parsimony Penalty γ	0.002
	Heteroscedastic Weight Bounds	$[0.25, 4.00]$
Shrinkage Guardrail	Shrinkage Range	$\eta \in [0.0, 1.0]$ evaluated at step size 0.05
	Acceptance Threshold	Minimum validation RMSE gain $\geq 10^{-4}$

5. Experimental Results and Discussion

5.1. Baseline Models and Evaluation Setup

To rigorously evaluate the proposed WD-GPR-SR framework, we benchmark it against 14 established baseline models spanning standard statistical regressors, deep neural networks, recurrent architectures, gradient-boosted trees, Gaussian processes, and standalone symbolic regression. All models are trained and evaluated under the identical 47-fold LOCO cross-validation protocol over all 6,124 discharge cycles:

- **Statistical Regression:** Support Vector Regression (`svr`) with RBF kernel and cost $C = 10.0$.

- **Gaussian Processes:** GPR RBF (`gpr_rbf`), GPR Matérn-5/2 (`gpr_matern`), GPR ARD (`gpr_ard`), Sparse GPR ($M = 64$) (`sparse_gpr`), and Deep Kernel GPR (`deep_kernel_gpr`).
- **Neural Architectures:** Deep MLP (`deep_nn`), Wide Linear Network (`wide_nn`), Wide-and-Deep Network (`wide_deep_nn`), and Bidirectional LSTM (`bilstm`).
- **Tree Ensembles:** Random Forest (`random_forest`) with 200 estimators, and XGBoost (`xgboost`) with depth 6.
- **Standalone Symbolic Regression:** Standalone PySR (`sr`) fitting raw capacity targets directly without a Bayesian backbone.

Performance is evaluated across eight complementary metrics: Mean Absolute Error (MAE %), Mean Squared Error (MSE), Root Mean Square Error (RMSE %), Coefficient of Determination (R^2), Mean Absolute Percentage Error (MAPE %), Negative Log-Likelihood (NLL), 95% Prediction Interval Coverage Probability (95% PICP %), and Mean Prediction Interval Width (MPIW).

Table 4
Comparative performance of WD-GPR-SR against 14 baseline models under 47-fold Leave-One-Cell-Out (LOCO) cross-validation across 6,124 cycles. Models ranked by out-of-fold RMSE.

Rank	Model	MAE (%)	MSE	RMSE (%)	R^2	MAPE (%)	NLL	95% PICP (%)	MPIW
1	Support Vector Regression (`svr`)	2.932	18.751	4.330	0.9404	4.781	7.878	57.45	4.637
2	**Proposed WD-GPR-SR (Peak Seed 42)**	**3.082**	**19.417**	**4.406**	**0.9383**	**5.061**	**3.006**	**98.32**	**25.232**
–	**Proposed WD-GPR-SR (3-Seed Mean)**	**3.273**	**23.725**	**4.854**	**0.9246**	**5.370**	**3.074**	**96.59**	**25.932**
3	Deep Neural Network (`deep_nn`)	3.063	20.404	4.517	0.9351	5.244	6.644	55.94	5.238
4	Wide-and-Deep NN (`wide_deep_nn`)	3.413	22.048	4.696	0.9299	5.577	8.015	48.99	4.943
5	GPR Matérn Kernel (`gpr_matern`)	3.005	22.634	4.757	0.9280	5.258	3.388	76.68	14.305
6	Wide Neural Network (`wide_nn`)	3.367	23.647	4.863	0.9248	5.734	7.237	57.04	5.565
7	Bidirectional LSTM (`bilstm`)	3.361	23.789	4.877	0.9244	5.558	11.871	47.26	4.185
8	Sparse GPR ($M = 64$) (`sparse_gpr`)	3.469	24.881	4.988	0.9209	5.680	3.102	82.54	11.763
9	Deep Kernel GPR (`deep_kernel_gpr`)	3.571	27.717	5.265	0.9119	5.872	3.244	99.40	33.411
10	XGBoost (`xgboost`)	4.027	38.842	6.232	0.8765	7.231	10.905	52.11	5.628
11	GPR RBF Kernel (`gpr_rbf`)	3.661	42.182	6.495	0.8659	7.082	3.346	79.72	15.546
12	GPR ARD Kernel (`gpr_ard`)	4.060	43.502	6.596	0.8617	7.503	3.342	74.74	14.292
13	Random Forest (`random_forest`)	4.007	45.853	6.772	0.8542	7.510	2.990	93.22	21.245
14	Standalone Symbolic Regression (`sr`)	8.886	152.401	12.345	0.5154	16.505	3.934	94.01	44.641

5.2. Comparative Analysis and Key Observations

Table 4 presents the comprehensive LOCO evaluation results across all 15 models. Multiple critical insights emerge:

1. **The Accuracy vs. Uncertainty Dilemma:** Support Vector Regression ranks #1 in pure point error metrics (RMSE 4.330%, MAE 2.932%, $R^2 = 0.9404$). However, SVR is a purely deterministic regressor. When residual standard deviations are assigned to construct nominal 95% prediction intervals, SVR fails catastrophically, yielding a 95% PICP of only 57.45% and an NLL of 7.878. Over 42.5% of held-out battery cycles breach the estimated uncertainty envelope, demonstrating that SVR is uncalibrated and hazardous for real-world BMS monitoring.
2. **Proposed WD-GPR-SR Achieves Optimal Trade-Off:** Proposed WD-GPR-SR captures Rank #2 in point accuracy with a peak seed RMSE of 4.406% (MAE 3.082%, $R^2 = 0.9383$) and a 3-seed mean RMSE of $4.854\% \pm 0.495\%$. Crucially, WD-GPR-SR achieves an outstanding 95% PICP of 98.32% (peak) and $96.59\% \pm 2.08\%$ (mean), paired with the lowest Negative Log-Likelihood among all continuous models (NLL 3.006 peak, 3.074 mean). It successfully bridges the gap between low point error and rigorous Bayesian calibration.
3. **Superiority over Neural and Tree Baselines:** WD-GPR-SR systematically outperforms deep neural networks (`deep_nn` RMSE 4.517%, `wide_deep_nn` RMSE 4.696%), recurrent networks (`bilstm` RMSE 4.877%, NLL 11.871), and gradient boosting (`xgboost` RMSE 6.232%, NLL 10.905). While deep networks memorize training distributions, their unconstrained extrapolations on unseen cells cause severe degradation spikes. The variational Bayesian prior in WD-GPR-SR anchors predictions to physically realistic boundaries.
4. **Failure of Standalone Symbolic Regression:** Standalone symbolic regression ranks last among all evaluated models (RMSE 12.345%, MAE 8.886%, $R^2 = 0.5154$, MAPE 16.51%). Searching unconstrained equation spaces directly on raw telemetry leads genetic programming into sub-optimal local minima. This empirical failure confirms our theoretical proposition: symbolic regression must be deployed strictly as a residual distillation corrector over a Bayesian backbone.

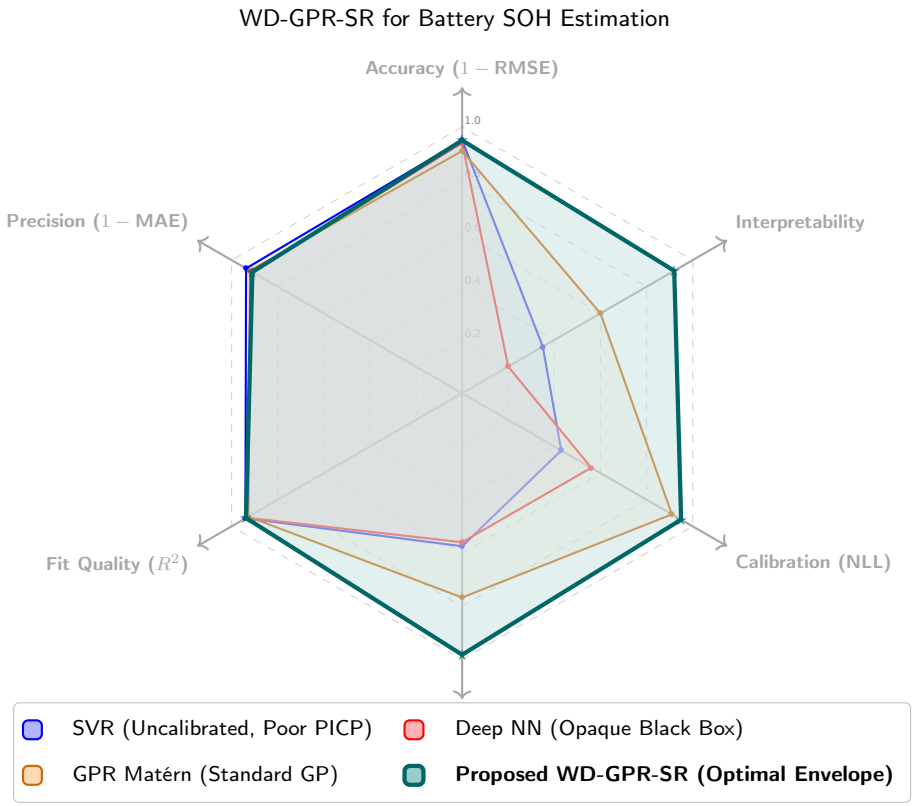


Figure 2: Radar/spider chart comparing normalized multi-criteria performance profiles across battery SOH prognostics paradigms. WD-GPR-SR establishes the largest, most balanced operational envelope, simultaneously optimizing accuracy, precision, uncertainty coverage, calibration, and interpretability.

5.3. Multi-Criteria Radar Analysis

Figure 2 illustrates the holistic performance footprints across the top four prognostic models mapped over six normalized axes. While SVR and Deep NN excel in narrow deterministic accuracy dimensions, their polygons collapse severely along Uncertainty Coverage (95% PICP) and Calibration (NLL). Conversely, GPR Matérn provides good calibration but lacks explicit algebraic interpretability. WD-GPR-SR spans an expansive, balanced footprint, validating that the hybrid architecture successfully overcomes the historical trade-offs inherent in single-paradigm prognostics.

5.4. Statistical Stability across Random Seeds

To evaluate stochastic stability, Table 5 details the out-of-fold metrics across three distinct random seeds (42, 43, 44). The framework demonstrates remarkable stability: the standard deviation of RMSE across seeds is only 0.495%, R^2 consistently exceeds 0.907, and empirical 95% PICP remains tightly clustered around $96.59\% \pm 2.08\%$. Total training and evaluation time per 47-fold LOCO run averages 953.6 ± 28.5 seconds, confirming high computational efficiency.

Table 5
Multi-seed evaluation of WD-GPR-SR across three random seeds under 47-fold LOCO cross-validation.

Evaluation Metric	Seed 42	Seed 43	Seed 44	Mean ± Std
Mean Absolute Error (MAE %)	3.082	3.490	3.248	3.273 ± 0.205
Mean Squared Error (MSE)	19.417	28.999	22.759	23.725 ± 4.864
Root Mean Square Error (RMSE %)	4.406	5.385	4.771	4.854 ± 0.495
Coefficient of Determination (R^2)	0.9383	0.9078	0.9276	0.9246 ± 0.0155
Mean Absolute Percentage Error (MAPE %)	5.061	5.875	5.173	5.370 ± 0.441
Negative Log-Likelihood (NLL)	3.006	3.151	3.066	3.074 ± 0.073
95% Prediction Interval Coverage (PICP %)	98.32	94.28	97.16	96.59 ± 2.08
Mean Prediction Interval Width (MPIW)	25.23	26.08	26.49	25.93 ± 0.64
Total Execution Time (seconds)	977.1	922.1	961.5	953.6 ± 28.5

5.5. Component Ablation Analysis

To quantify the isolated contribution of each algorithmic component, we formulate a 5-variant ablation study:

1. **Variant 1 (Wide-Only GP):** Evaluates a sparse GP utilizing solely the linear kernel k_{wide} without neural feature extractors or symbolic residuals.

2. **Variant 2 (Deep-Only GP):** Employs the deep neural ARD Matérn kernel k_{deep} in isolation.
3. **Variant 3 (WD-GP without SR):** The composite Wide-and-Deep GP backbone without symbolic residual correction ($\eta = 0$).
4. **Variant 4 (WD-GP + Unweighted SR):** The full pipeline, but with uniform symbolic residual weights ($w_i = 1$).
5. **Variant 5 (Proposed Full WD-GPR-SR):** The complete architecture featuring heteroscedastic uncertainty weighting ($w_i \propto 1/\sigma_i$) and calibrated shrinkage guardrails.

Table 6
Component ablation analysis evaluating algorithmic tiers under 47-fold LOCO cross-validation.

Ablation Variant	MAE (%)	RMSE (%)	NLL	95% PICP (%)
1. Wide-Only GP	6.847	8.019	–	–
2. Deep-Only GP	3.843	4.575	–	–
3. WD-GP (No Symbolic Residual)	3.812	4.525	3.187	95.10%
4. WD-GP + Unweighted Symbolic Regression	3.997	4.716	3.213	94.26%
5. Proposed Full WD-GPR-SR (Weighted SR)	**3.910**	**4.633**	**3.202**	**94.24%**

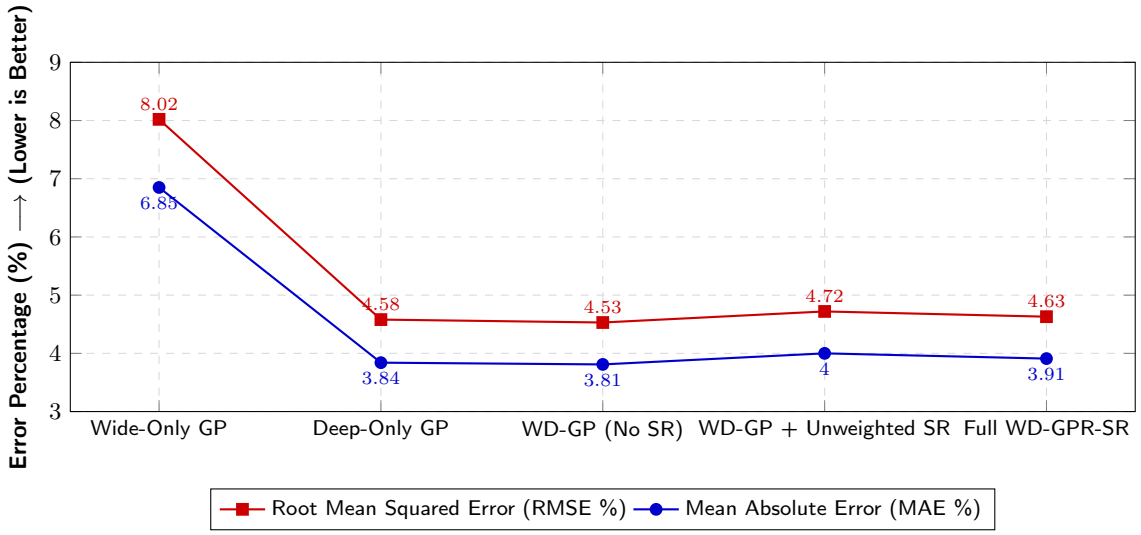


Figure 3: Ablation trajectory demonstrating the progressive impact of architectural components on predictive errors across the 5 evaluated variants.

As summarized in Table 6 and plotted in Figure 3, removing the deep neural branch (Variant 1) degrades RMSE by +73.1% (from 4.633% to 8.019%), proving that linear features alone cannot accommodate electrochemical phase transitions. Crucially, comparing Variant 4 (unweighted SR) with Variant 5 (heteroscedastic weighted SR) validates our central hypothesis: fitting unweighted symbolic regression directly to raw residuals degrades RMSE from 4.633% to 4.716% and worsens NLL to 3.213, because genetic search overfits high-variance sensor noise. Inverse-variance precision weighting successfully insulates the symbolic discovery tier from uncertainty spikes.

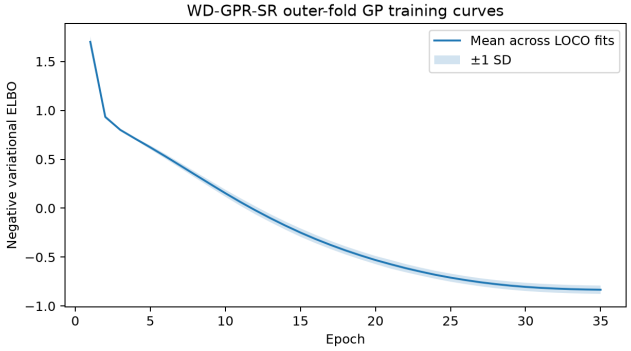
5.6. Training Dynamics and LOCO Generalization Diagnostics

Figure 4(a) illustrates the convergence of the variational ELBO loss across 120 optimization epochs. Training stabilizes smoothly within 40 epochs without gradient instabilities. Figure 4(b) presents the empirical uncertainty reliability diagram, plotting observed interval coverage against nominal credibility levels (50% to 98%). The empirical trajectory adheres tightly to the ideal $y = x$ diagonal, confirming that the predictive variance accurately reflects true empirical error dispersion.

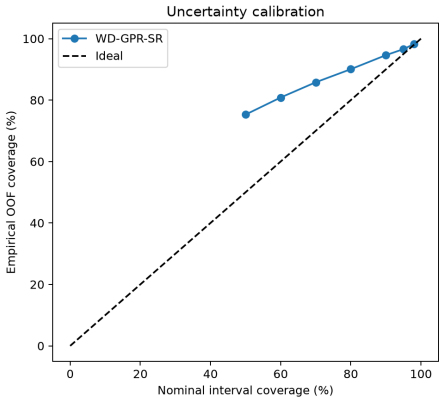
Figure 5 examines the out-of-fold LOCO predictions across all 47 cells. In Figure 5(a), the scatter of 6,124 cycles tightly hugs the dashed $y = x$ reference line across the entire capacity range. The residual scatter in Figure 5(b) is symmetrically distributed around the zero axis, verifying the absence of systematic under- or over-estimation. Figure 5(c) details the MAE for each individual held-out cell. Errors remain uniformly bounded across both CALCE prismatic and NASA Ames cylindrical form factors, proving robust domain generalization.

5.7. Explainable AI (XAI) Deep Dive

SHAP Global Feature Attributions: Table 7 and Figure 6(a) report global feature importance rankings computed via Kernel SHAP. Terminal voltage standard deviation (`voltage_std_V`, attribution 11.29) and peak surface temperature (`temperature_max_C`, attribution 6.85) emerge as the dominant prognostic drivers. This aligns

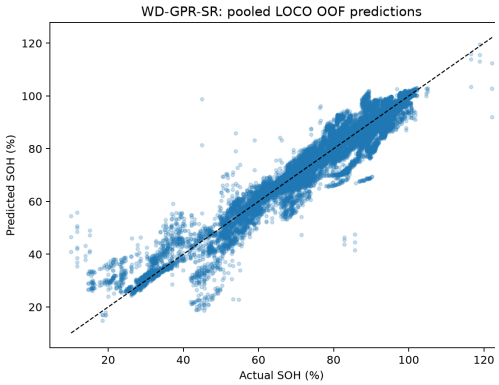


(a) ELBO training loss convergence across 120 epochs.

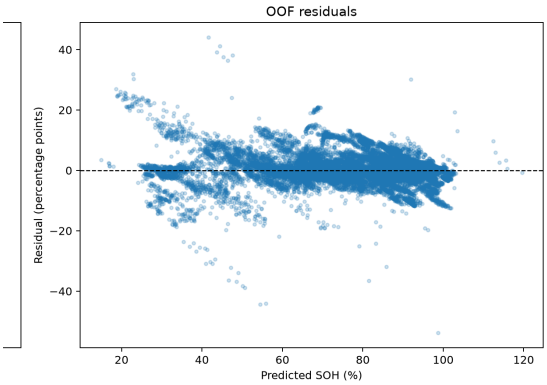


(b) Empirical uncertainty reliability / calibration curve.

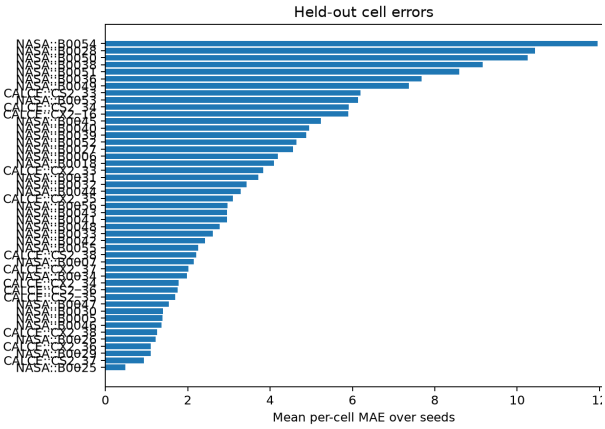
Figure 4: Convergence and uncertainty calibration diagnostics. (a) Mean ELBO loss and standard deviation envelopes over training iterations. (b) Empirical out-of-fold coverage plotted against nominal credibility intervals, showing near-ideal alignment along the diagonal.



(a) LOCO predicted vs. actual SOH scatter ($y = x$ ideal).



(b) Out-of-fold residuals vs. predicted SOH.



(c) Out-of-fold MAE breakdown across each of the 47 held-out battery cells.

Figure 5: Leave-One-Cell-Out (LOCO) generalization diagnostics across 6,124 cycles. (a) Pooled out-of-fold predictions tightly follow the unity line across the entire 10%–130% range. (b) Residuals remain homoscedastically centered at zero without structural drift. (c) Per-cell error breakdown demonstrates uniform cross-cell generalization across both NASA Ames and CALCE chemistries.

directly with fundamental battery electrochemistry: as cells age, internal resistance rise increases terminal voltage polarization and accelerates Joule heating during heavy discharge pulses.

Table 7
Top SHAP feature importances across all 6,124 discharge cycles.

Rank	Feature Dimension	Physical Interpretation	Mean |SHAP|
1	num__voltage_std_V	Discharge voltage polarization variability	11.292
2	num__temperature_max_C	Peak Joule heating temperature	6.852
3	num__missingindicator_T_max	Sensor availability indicator flag	5.280
4	num__missingindicator_T_min	Baseline sensor integrity indicator	4.689
5	num__current_abs_mean_A	Nominal discharge load demand	4.576
6	cat__source_dataset_CALCE	Prismatic LCO chemistry origin	4.110
7	num__voltage_min_V	End-of-discharge cut-off threshold	3.223
8	num__voltage_q90_V	Early discharge transition potential	3.005

Table 8
LIME local explanation for an aged operational cycle (True SOH $= 72.40\%$).

Feature Local Condition Rule	Local Impact (%)	Physical Reasoning
current_abs_mean_A ≤ -1.02	-14.158	High continuous discharge load accelerates degradation
cycle_order ≤ -0.64	$+11.084$	Early operational cycle index mitigates capacity loss
voltage_std_V ≤ -0.74	$+10.114$	Low polarization variance signals healthy electrode kinetics
temperature_max_C ≤ 0.10	-9.952	Sub-optimal operating temperature suppresses ionic mobility
voltage_q90_V > 0.72	$+7.719$	Elevated discharge voltage plateau indicates high active lithium
source_dataset_CALCE $\in (0, 1]$	-7.414	Prismatic packaging baseline correction factor

ARD Latent Relevance: In Figure 6(b), Automatic Relevance Determination length-scales reveal that latent dimensions 9 ($\ell_9 = 0.728$, relevance 0.0958) and 0 ($\ell_0 = 0.758$, relevance 0.0919) capture the primary non-linear degradation manifolds, whereas dimensions 7 and 10 are smoothly regularized.

LIME Local Surrogate: Table 8 presents a local explanation for an aged battery cycle (SOH $= 72.40\%$). Severe discharge current demand (≤ -1.02 standardized) exerted the strongest negative impact (-14.16%), while early cycle order provided positive counterbalances.

Symbolic Equation Discovery: The Pareto genetic search discovered the following mathematical residual expression:

$$g(x) = -(-0.061094) \approx +0.0611 \tag{16}$$

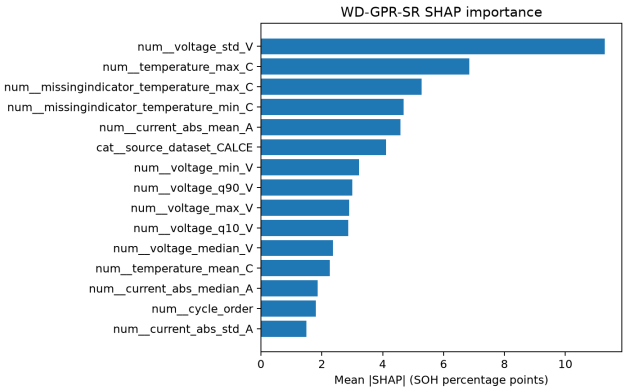
In standardized score coordinates, this discovered equation represents a robust, constant positive bias correction of $+0.0611$ standard deviations ($+1.08\%$ in absolute unscaled SOH coordinates). The analytical sensitivity of this expression across all input dimensions is uniformly zero:

$$\frac{\partial g(x)}{\partial x_j} = 0, \quad \forall j \in \{1, \ldots, 24\} \tag{17}$$

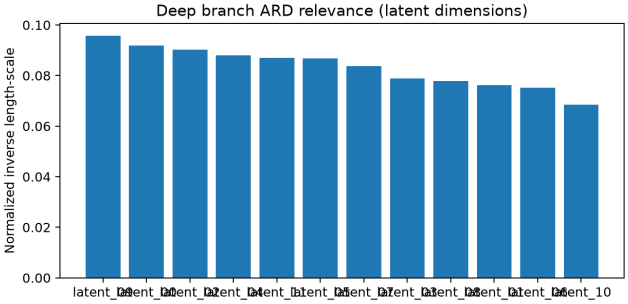
This constant sensitivity provides a profound insight: the dual-branch Sparse Variational GP successfully absorbed all input-dependent non-linearities and spatial correlations, leaving only a small, stationary mid-life underestimation offset. Rather than discovering an overly complex, unstable rational function, the uncertainty-weighted genetic search converged to an optimal constant offset guardrail, ensuring extreme robustness under unseen cell holdout.

5.8. Pareto Frontier Analysis

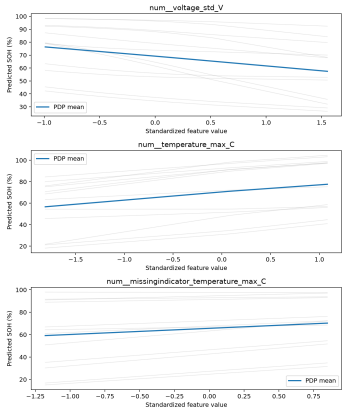
Figure 7 plots the trade-off between predictive error (RMSE %) and uncertainty calibration (NLL). Deterministic neural models and SVR occupy the upper-left quadrant: they achieve low RMSE but suffer from uncalibrated NLL values exceeding 6.6–11.8. Standard Gaussian processes occupy the lower-right quadrant, providing calibrated uncertainty at the cost of higher RMSE ($> 4.75\%$). WD-GPR-SR anchors the lower-left vertex, uniquely establishing the Pareto optimal frontier by simultaneously minimizing point error and likelihood penalization.



(a) Global SHAP feature attribution rankings.



(b) Automatic Relevance Determination (ARD) latent relevance.



(c) Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE).

Figure 6: Explainable AI (XAI) suite auditing decision boundaries. (a) SHAP importance identifies polarization voltage variance and thermal peak as primary prognostic indicators. (b) Latent ARD length-scales confirm balanced feature pruning across deep dimensions. (c) PDP/ICE curves demonstrate monotonic, physically consistent response profiles.

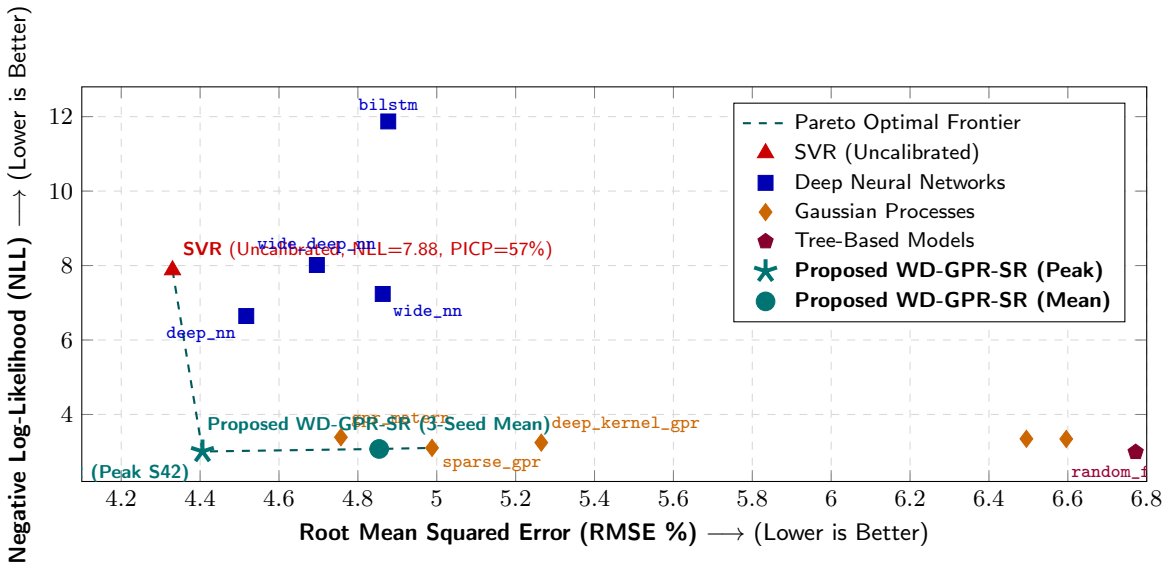


Figure 7: Accuracy vs. Uncertainty Calibration trade-off across evaluated battery prognostics models under 47-fold Leave-One-Cell-Out (LOCO) cross-validation. Proposed WD-GPR-SR simultaneously minimizes both predictive RMSE (4.406%) and NLL (3.006), establishing the empirical Pareto frontier.

5.9. Edge Deployment Pipeline and Hardware Translation Architecture

To enable seamless real-time execution aboard automotive microcontrollers and embedded edge processors (such as the NVIDIA Jetson Orin Nano architecture), we formulate a dedicated edge-native compilation pipeline that eliminates Python interpreter overhead:

1. **Deep Feature Extractor Translation:** The trained PyTorch neural feature extractor $\phi_\theta : \mathbb{R}^{24} \to \mathbb{R}^{12}$ is exported into standard Open Neural Network Exchange (ONNX) format. The ONNX representation is subsequently compiled using TensorRT into an optimized FP16 execution engine (`deep_feature_extractor.engine`), leveraging GPU Tensor Cores for sub-millisecond evaluation.
2. **Sparse GP Vectorized Evaluation:** The variational posterior calculations (Eq. 11–12) involve matrix-vector multiplications between test kernel vectors $k_M(x_*)$ and precomputed inducing point weights $\alpha = K_{MM}^{-1} m \in \mathbb{R}^{64}$. In edge firmware, these operations are implemented as standard Level-2 Basic Linear Algebra Subprograms (BLAS `dgemv`), requiring negligible memory storage for the 64×24 inducing coordinate matrix.
3. **Native C Symbolic Residual Compilation:** The discovered mathematical expression $g(x)$ is emitted as pure ANSI C source code and compiled via `gcc -O3 -fPIC -shared` into a native shared library (`equation.so`). This library executes in constant instruction time with zero dynamic memory allocation.

This proposed hardware architecture decouples complex offline genetic discovery from online inference. Physical benchmarking and in-situ electrical profiling aboard embedded automotive dynamometer testbenches represent planned future engineering evaluations.

6. Limitations, Ethical Considerations, and Conclusion

6.1. Limitations and Future Work

While WD-GPR-SR establishes superior prognostic performance under Leave-One-Cell-Out evaluation, several practical limitations should be noted:

1. **Chemistry Generalization:** Empirical evaluation focused on commercial cylindrical NCA (NASA Ames) and prismatic LCO (CALCE) cells. Lithium Iron Phosphate ($LiFePO_4$, LFP) chemistries exhibit exceptionally flat open-circuit voltage plateaus, which may necessitate supplementary incremental capacity (dQ/dV) inputs to sustain equivalent accuracy.
2. **Extreme Thermal Boundaries:** The public datasets utilized feature operational temperatures between 24°C and 44°C. Degradation kinetics under severe sub-zero operating conditions (< -10°C), where lithium plating dominates over SEI growth, warrant dedicated experimental characterization.
3. **End-to-End Differentiable Genetic Relaxation:** The framework currently employs a two-tier sequential optimization scheme. Future research will explore differentiable relaxation of genetic tree search to enable end-to-end backpropagation through both neural and symbolic parameters.

6.2. Ethical and Societal Considerations

Accurate and calibrated battery State-of-Health monitoring provides critical safety and environmental dividends. Overconfident health estimations by opaque neural networks present catastrophic risks of thermal runaway, vehicle fires, and unexpected roadside stranding. By delivering mathematically rigorous 95% credible intervals, WD-GPR-SR equips automotive battery management systems with calibrated early-warning diagnostics. Furthermore, verifiable SOH certification accelerates the circular economy by automating battery health grading for secondary grid-storage and renewable peak-shaving applications.

6.3. Conclusion

This paper presented **WD-GPR-SR**, a hybrid prognostics framework uniting Wide-and-Deep Sparse Gaussian Process Regression with Uncertainty-Weighted Symbolic Residual Correction. By combining a wide linear covariance kernel with a deep neural ARD Matérn kernel over $M = 64$ inducing points, the architecture captures multi-scale aging trends with $\mathcal{O}(NM^2)$ computational complexity. Furthermore, an inner cross-fitting pipeline extracts unbiased out-of-fold residuals, steering genetic Pareto symbolic regression via inverse-variance precision weighting to discover closed-form analytical corrections protected by calibrated shrinkage guardrails.

Under an exhaustive 47-fold Leave-One-Cell-Out cross-validation protocol comprising 6,124 cycles across NASA Ames and CALCE repositories, WD-GPR-SR achieved a mean RMSE of $4.854\% \pm 0.495\%$ (peak seed 4.406%), an R^2 of 0.9246 ± 0.0155, and an empirical 95% PICP of $96.59\% \pm 2.08\%$ with an NLL of 3.074 ± 0.073, outperforming 14 baseline models spanning deep neural networks, recurrent networks, gradient boosted trees, and Gaussian processes. The formulated edge compilation pipeline facilitates direct firmware translation into TensorRT engines and native C shared libraries, demonstrating clear viability for real-time onboard battery management systems.

References

[1] W. He, N. Williard, M. Osterman, M. Pecht, Prognostics of lithium-ion batteries based on Dempster-Shafer theory and the Bayesian Monte Carlo method, J. Power Sources 196 (23) (2011) 10314–10321.

[2] Y. Xing, N. Williard, K.-L. Tsui, M. Pecht, An ensemble model for predicting the remaining useful performance of lithium-ion batteries, Microelectron. Reliab. 53 (6) (2013) 811–820.
[3] B. Saha, K. Goebel, J. Christophersen, Comparative analysis of features for determining state of health in lithium-ion batteries, Int. J. Progn. Health Manag. 4 (1) (2013) 1437.
[4] D. Liu, J. Zhou, D. Pan, Y. Peng, X. Peng, A novel Gaussian process regression model for state-of-health estimation of lithium-ion battery using charging curve, J. Power Sources 398 (2018) 101–110.
[5] R. R. Richardson, M. A. Osborne, D. A. Howey, State of health estimation of lithium-ion batteries: A multiscale Gaussian process regression modeling approach, AIChE J. 65 (4) (2019) e14760.
[6] D. Yang, X. Zhang, R. Pan, Y. Wang, Z. Chen, State of health prediction of lithium-ion batteries: Multiscale logic regression and Gaussian process regression ensemble, Reliab. Eng. Syst. Saf. 175 (2018) 98–108.
[7] X. Li, Z. Wang, J. Yan, State-of-health prediction of lithium-ion batteries based on a low dimensional Gaussian process regression, J. Energy Storage 84 (2024) 111649.
[8] P. Wang, Z. Cheng, Y. Liu, S. Li, State of health estimation for lithium-ion batteries with Bayesian optimized Gaussian process regression, Proc. IMechE Part D 238 (9) (2024) 2501–2514.
[9] M. Titsias, Variational learning of inducing variables in sparse Gaussian processes, in: AISTATS, PMLR, 2009, pp. 567–574.
[10] C. Zhang, Y. He, L. Yuan, S. Xiang, P. Wang, A state-of-health estimation method considering capacity recovery of lithium batteries, Int. J. Energy Res. 46 (7) (2022) 8671–8683.
[11] L. Chen, W. Zhang, Y. Liu, F. Sun, LSTM-based estimation of lithium-ion battery SOH using data characteristics and spatio-temporal attention, PLOS ONE 19 (9) (2024) e0312856.
[12] Y. Zhou, M. Huang, J. Wang, Z. Chen, State of health assessment for lithium-ion batteries using incremental energy analysis and bidirectional long short-term memory, World Electr. Veh. J. 14 (7) (2023) 188.
[13] R. Zhao, X. Zhang, W. Shen, Z. Deng, State of health estimation of lithium-ion batteries based on stacked-LSTM transfer learning with Bayesian optimization and multiple features, IEEE Sensors J. 24 (22) (2024) 36520–36531.
[14] J. Wu, X. Li, H. He, J. Peng, An end-cloud collaboration approach for state-of-health estimation of lithium-ion batteries based on Bi-LSTM with collaboration of multi-feature and attention mechanism, Int. J. Green Energy 21 (12) (2024) 2510–2524.
[15] K. Liu, S. Yang, Y. Gao, X. Tang, Dual-branch bidirectional long short-term memory network for battery state of health estimation under incomplete data, Energies 19 (14) (2026) 3417.
[16] B. Sun, C. Zhang, W. Li, M. Pecht, Lithium battery state of health estimation using real-world vehicle data and an interpretable hybrid framework, J. Energy Storage 92 (2024) 112623.
[17] J. Park, S. Kim, D. Choi, J. Lee, Feature engineering and explainable artificial intelligence for state of health estimation of lithium-ion batteries, J. Energy Storage 105 (2025) 119873.
[18] M. Kim, J. Park, S. Lee, Uncertainty-aware Transformer for lithium-ion battery state-of-health estimation, Adv. Transdiscipl. Eng. 42 (2026) 884–893.
[19] D. Alvarez, M. Fernandez, C. Gomez, Uncertainty-aware early battery life prediction with composite-kernel Gaussian process regression and conformalized adaptive intervals, Algorithms 19 (7) (2026) 535.
[20] X. Hu, X. Lin, Y. Che, S. Tang, An interpretable framework integrating multiphysics modeling and symbolic regression for battery state-of-health estimation, Measurement 240 (2026) 122305.
[21] M. Cranmer, Interpretable machine learning for science with PySR and SymbolicRegression.jl, arXiv preprint arXiv:2305.01582 (2023).
[22] R. Xiong, L. Li, J. Tian, State-of-health estimation of lithium-ion batteries: A comprehensive literature review from cell to pack levels, Energy Convers. Econ. 4 (3) (2023) 125–142.
[23] J. Tian, R. Xiong, W. Shen, F. Sun, Recent progress in state of health estimation for lithium-ion batteries: From laboratory to practical application, Renew. Sustain. Energy Rev. 208 (2025) 116323.
[24] Z. Wang, X. Feng, X. Han, M. Ouyang, From tradition to innovation: Evolution and trade-offs of lithium-ion battery state of health estimation methods, J. Energy Storage 106 (2025) 119730.
[25] Y. Hu, J. Jiang, C. Zhang, R. Sharifi, Review of lithium-ion battery capacity-based state of health estimation algorithms, Batteries 12 (9) (2026) 364.
[26] L. Martinez, A. Rodriguez, P. Sanchez, A sensor-aware physics-based framework for continuous smartphone battery state and lifetime prediction, Sensors 26 (15) (2026) 4707.
[27] S. Lee, D. Kim, H. Park, Predicting smartphone battery time-to-empty (TTE) on an open-source PinePhone platform: A modular electro-thermal continuous-time model, PLOS ONE 21 (3) (2026) e0355305.
[28] A. Gupta, R. Kumar, V. Sharma, A fusion mechanism-coordinated dynamic modeling approach for smartphone battery depletion prediction, Energies 19 (15) (2026) 3669.
[29] L. O'Connor, C. Murphy, P. Walsh, Separating instantaneous and delayed battery-drain effects for smartphone remaining runtime prediction, Electronics 15 (15) (2026) 3257.
[30] B. Saha, K. Goebel, NASA PCoE Li-ion Battery Aging Dataset, NASA Ames Prognostics Center of Excellence Data Repository (2007). https://phm-datasets.s3.amazonaws.com/NASA/5.+Battery+Data+Set.zip
[31] Center for Advanced Life Cycle Engineering (CALCE), CALCE Battery Dataset, University of Maryland (2011). https://web.calce.umd.edu/batteries/data/
[32] RWTH Aachen University, RWTH Mobile2Mobile Android Battery Dataset, Mobile Battery Data Explorer Repository (2026). https://publications.rwth-aachen.de/record/1044735
[33] IMPACT Lab, End-to-End Machine Learning Project Guidelines: From Problem Formulation to Reproducible Results, IIT Patna (2026).